Task 2 - Embeddings and Semantic Search: Comparison Table

Model: sentence-transformers all-MiniLM-L6-v2 | 12 FAQ questions (HR, IT, Payroll) | cosine_similarity implemented with numpy

Query	Keyword result	Semantic result (top 3)
I cannot log in	No match found	1. [IT-1] How do I reset my password if I am locked out of my account? (cosine 0.51) 2. [IT-3] How can I request a new laptop or replacement equipment? (cosine 0.19) 3. [IT-4] Who do I contact when the printer or Wi-Fi stops working? (cosine 0.17)
when is salary credited	No match found	1. [PAY-1] On which date are monthly wages deposited into my bank account? (cosine 0.55) 2. [PAY-4] How are income tax deductions and reimbursements calculated? (cosine 0.35) 3. [HR-2] How many days of annual leave am I entitled to each year? (cosine 0.32)
work from home rules	1. [HR-1] What are the rules for working from home or remote work? (matched words: 3)	1. [HR-1] What are the rules for working from home or remote work? (cosine 0.77) 2. [HR-2] How many days of annual leave am I entitled to each year? (cosine 0.18) 3. [IT-2] How do I connect to the office VPN from outside? (cosine 0.13)

Queries where semantic search wins

'I cannot log in' (correct FAQ: IT-1): The query says 'log in' but the FAQ says 'password' and 'locked out'. They share no words, yet embeddings place both near the idea of an access/login problem.

'when is salary credited' (correct FAQ: PAY-1): The FAQ uses 'wages' and 'deposited' instead of 'salary' and 'credited'. Keyword search finds no common word; embeddings treat these synonyms as close in meaning.